e-cell
IIT BOMBAY

everseen

COMPUTER VISION AI HACKATHON

WasteSense AI

Predict Before Waste. Act Before Loss.

An AI-powered platform to predict unsold food inventory and recommend preventive actions before it becomes waste.



Problem Statement

Predicting the Unseen:
Pre-Visual Food Waste Mitigation

Team Name
Syntax Error

e-cell IIT BOMBAY | everseen

COMPUTER VISION AI HACKATHON

Team Members



Nishant Bobade

Team Leader

AI/ML & Backend Developer

LinkedIn: Click here



Vaishali Kumbhare

Frontend & UI/UX Developer

Frontend & UI/UX Developer

LinkedIn: Click here



Parv Bisen

Research & Data Analysis

Research & Data Analysis

LinkedIn: Click here



Sumit Badshah

Testing & Model Evaluation

Model Testing, Evaluation & QA

LinkedIn: Click here

Team Name

Syntax Error

Syntax Error

PROBLEM & APPROACH

COMPUTER VISION AI HACKATHON

e-cell IIT BOMBAY | everseen



Inventory
Available stock

↓



Expiry Tracking
Know what's at risk

↓



Demand Prediction
Anticipate future sales

↓



Smart Actions
Prevent waste, recover value

3 CORE CHALLENGES

1 

Demand Uncertainty

Daily sales fluctuate due to seasonality, promotions, weather, and local demand, making surplus difficult to anticipate.

2 

Expiry-Window Pressure

Near-expiry batches need timely action before they become unsellable, leading to potential waste and financial loss.

3 

Fragmented Decisions

Inventory management, markdowns, store transfers, and donations are often handled separately, resulting in delayed or suboptimal decisions.

OUR APPROACH

1 

Demand Forecasting

Use historical sales data when available, with a transparent weighted moving-average baseline and fallback demand inputs when historical data is limited or unavailable.

2 

Batch-Level Waste-Risk Assessment

Combine available quantity, forecast demand until expiry, remaining selling days, and uncertainty. Provide clear explanations for why each batch is flagged as at risk.

3 

Action Recommendation Engine

Compare markdown, store transfer, replenishment adjustment, eligible donation, or human inspection. Enforce expiry and food-safety constraints.

Our Goal | Predict Before Waste. **Act Before Loss.**



Predict Forecast demand →  **Prevent** Take timely action →  **Recover** Redistribute or sell with discount →  **Measure** Track impact and learn

e-cell
IIT BOMBAY

everseen

COMPUTER VISION AI HACKATHON

INNOVATION & SOLUTION

WHAT WE BUILT

WasteSense AI is a food-waste intelligence platform that forecasts demand, identifies inventory likely to remain unsold before expiry, and recommends practical actions to prevent waste.

OUR SOLUTION

1. **Demand Forecasting**
   Uses historical sales when available, with a transparent weighted moving-average baseline and fallback demand inputs.



2. **Batch-Level Waste-Risk Scoring**
   Estimates demand until expiry, expected surplus, remaining selling days, and financial exposure, with clear reasons for each alert.



3. **Action Recommendation Engine**
   Compares markdowns, store transfers, replenishment changes, eligible donations, and inspection while enforcing expiry and food-safety constraints.



KEY PERFORMANCE TAKEAWAYS



Forecasting Readiness

- Weighted moving-average baseline
- Evaluate with MAE · RMSE · WAPE
  (Evaluation measures)



Explainable Risk

- Batch-level surplus estimates
- Expiry window + demand uncertainty



Actionable Prevention

- Discount · Transfer · Donate
- Track estimated vs. recorded outcomes

Predict Before Waste. Act Before Loss.

Team Name
Syntax Error

Syntax Error

TECHNICAL APPROACH

COMPUTER VISION AI HACKATHON

e-cell IIT BOMBAY | everseen

ARCHITECTURE FLOW



Inventory + Sales + Expiry Data
Current stock, sales history, expiry dates

Data Validation & Batch Preparation
Clean, merge and prepare time-series data

Forecasting Engine

- **Weighted Moving Average** — Forecast future demand
- **Demand & Expiry Window** — Analyze demand till expiry
- **Surplus Estimation** — Predict likely unsold quantity

Waste-Risk Scoring
Estimate risk of each item/batch becoming waste

Action Recommendation Engine
Suggest best action based on risk, stock and context

- **Dashboard & Alerts** — Visual insights, notifications
- **Markdown / Transfer / Donation / Inspection** — Practical actions to prevent waste

EXPLAINABLE RISK MODEL



Expected Surplus = max(0, Available Units − Forecast Demand Until Expiry)

Waste-Risk Score (0 – 100)
A heuristic score based on multiple factors (not a calibrated probability).

- **Surplus Ratio** — Expected surplus relative to stock — Weight: Configurable
- **Days Until Expiry** — Closer expiry increases risk — Weight: Configurable
- **Demand Uncertainty** — Variability in demand forecast — Weight: Configurable

TECH STACK / MODULES

Frontend	User interface, dashboard and visualizations	React, TypeScript, Vite
Inventory Data	Import and manage product & sales data	CSV Import, LocalStorage
Forecasting	Demand prediction and surplus estimation	Weighted Moving Average, Baseline Comparison
Risk Engine	Batch-level risk scoring with explainable factors	Rule-based Heuristics, Feature Analysis
Decision Engine	Recommend best action to prevent waste	Markdown, Transfer, Replenishment, Donation
Evaluation	Validate forecasting and risk estimation	MAE, RMSE, WAPE, Held-Out Validation

Predict Before Waste. **Act Before Loss.**

Team Syntax Error

Syntax Error

IMPACT & BENEFITS

COMPUTER VISION AI HACKATHON

e-cell IIT BOMBAY | everseen



1. ENVIRONMENTAL & SOCIAL IMPACT

Less waste, more food recovered

Prevent avoidable waste
Flag potential surplus before expiry using demand and inventory insights.

Redirect eligible surplus
Recommend donation when food is safe and feasible.

Improve availability
Help stores act before shelves and batches become a problem.

2. ECONOMIC & OPERATIONAL BENEFITS

Better decisions, less loss

Recover value
Compare markdowns, transfers and donation options.

Plan inventory better
Use demand forecasts to reduce over-ordering.

Prioritize staff action
Rank batches by urgency and expected surplus.

3. TRUST, SAFETY & MEASUREMENT

Explainable and accountable

Explain each alert
Show demand, expiry and surplus factors.

Respect food-safety rules
Expired or quarantined items are not sale/donation candidates.

Measure real outcomes
Separate estimates, simulations and verified results.

Predict Before Waste. **Act Before Loss.**

Team **Syntax Error**

Syntax Error

FEASIBILITY & VIABILITY

COMPUTER VISION AI HACKATHON

e-cell IIT BOMBAY | everseen

FEASIBILITY

Deployable MVP
React, TypeScript and Vite; core forecasting runs in the browser.

Accessible data
CSV inventory imports, public retail datasets where suitable, and clearly labelled demo data.

Transparent baseline
Weighted moving-average demand forecast and explicit expected-surplus calculation.

Low infrastructure needs
Static Netlify deployment; no paid API or backend required for the core demo.

Phased development
Start with explainable rules; add validated ML and hosted storage in later stages.

VIABILITY
FOR RETAILERS

Earlier intervention
Flag likely surplus before expiry using available stock, demand and remaining selling days.

Practical options
Compare markdown, transfer and replenishment adjustments.

Operational value
Prioritize batches and track estimated value at risk; prove outcomes in a pilot.

VIABILITY
FOR FOOD-RECOVERY PARTNERS

Eligible surplus matching
Identify suitable, unexpired batches for donation review.

Safer coordination
Record partner preferences, quantity and pickup window.

Responsible handling
Exclude expired or quarantined products; require human confirmation where safety is uncertain.

Predict Forecast demand → **Prevent** Take timely action → **Recover** Redistribute or donate → **Measure** Track impact and learn

Predict Before Waste. **Act Before Loss.**

Team Syntax Error

Syntax Error

RESEARCH & REFERENCES

COMPUTER VISION AI HACKATHON

e-cell IIT BOMBAY | everseen

ACADEMIC PAPERS & INDUSTRY REFERENCES

Key literature, datasets and real-world practices that guide our solution

1. **Food Waste Forecasting using Machine Learning**
   (Rahman et al., 2021)
   Explores ML models for perishable food demand forecasting in retail and restaurants, showing significant reduction in food waste.

2. **Dynamic Pricing for Perishable Products**
   (Caro & Gallien, 2007)
   Introduces dynamic markdown strategies to reduce food waste while maximizing revenue for retailers.

3. **Food Recovery and Redistribution Models**
   (Gunders, 2012)
   Analyzes effective food donation and redistribution systems to bridge surplus food with communities in need.

4. **Retail Demand Forecasting Datasets**
   (M5 Forecasting – Kaggle, 2020)
   Widely used dataset containing retail sales, calendar and price data for demand forecasting and inventory optimization.

5. **Food Safety and Donation Guidelines**
   (FAO, 2019)
   Provides global standards and safety guidelines for food redistribution and donation to ensure safe consumption.

DATA SOURCE NOTE
M5 Forecasting (Kaggle) dataset contains retail sales, calendar, and price data. Useful for demand forecasting, but does not include food-expiry labels.

MARKET RESEARCH & OPPORTUNITY

Estimated opportunity for surplus food management solutions



Key Growth Drivers

- Rising food waste awareness
- Sustainability regulations
- Growing adoption of AI in retail
- Increased food donation networks

Target Users

Supermarkets & Grocery Chains | Restaurants & Food Service | NGOs & Food Banks

Impact Alignment

- Supports UN SDG 12 (Responsible Consumption)
- Reduces landfill waste
- Improves food accessibility for underprivileged communities

Project Link — View Live Demo | Click here

GitHub Repo — Source Code | Click here

Predict Before Waste. **Act Before Loss.**